

Optimization of Small-Signal Model of GaN HEMT by Using Evolutionary Algorithms

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Abstract—In this letter, a new evolutionary algorithm-based method for the optimization of intrinsic elements of small-signal model (SSM) of GaN HEMT devices is presented. The method uses a unique search space exploration strategy for evolutionary algorithms to obtain an optimized compact SSM from the extracted parameter and measured S-parameters. The validity of the method is verified by comparing the measured S-parameter data of a $2 \times 0.1 \times 50 \mu\text{m}^2$ GaN/Si HEMT and a $4 \times 0.1 \times 75 \mu\text{m}^2$ GaN/SiC HEMT in the frequency range of 1 to 30 GHz. The modeled data and measured data are in good agreement.

Index Terms—Evolutionary algorithms, GaN HEMT, optimization.

I. INTRODUCTION

IN ORDER to design active circuits in the microwave frequency range, a necessary prerequisite is to have an accurate model of the implemented active device. Compact small signal models (SSMs) have become popular in recent past due to the ease of modeling, dynamic range and accuracy, and a good compromise between physics-based and tabular models. Moreover, SSMs are often used as a building block for large-signal modeling of high-power nonlinear devices in the form of multibias SSM [1]. An SSM involves the extraction and optimization of equivalent circuit elements of both extrinsic and intrinsic nature for a given device.

Different parameter extraction techniques have been proposed by Dambrine *et al.* [2], Berroth and Bosch [3], and Rorsman *et al.* [4]. These techniques primarily rely on analytical expressions and S-parameters measurements at the bias point of interest. These methods have limitation in the form of measurement noise and package parasitics at higher frequency affecting adversely the parameter extraction. Consequently, in order to overcome such limitations, Wurtz [5] proposed the method of deembedding the bias-independent extrinsic circuit elements and then using an optimization algorithm to arrive at an accurate intrinsic SSM. An SSM of different transistors operating in microwave and millimeter-wave frequency ranges has been determined in [6] using a similar

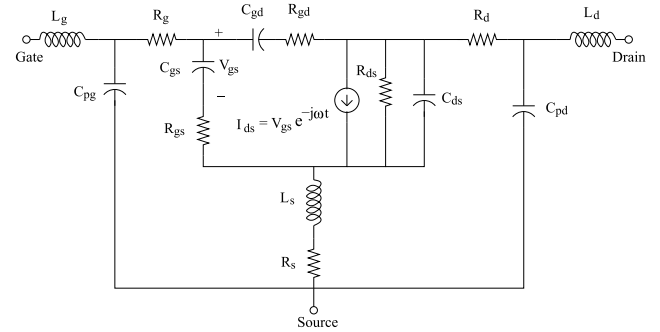


Fig. 1. SSM for GaN/Si and GaN/SiC HEMT.

method but a different optimization algorithm such as downhill simplex algorithm. A similar SSM modeling approach has been adopted in [7] and [8], where the method of least squares fitting and particle swarm optimization (PSO) has been considered as an optimization tool for SSM of GaN HEMT devices. In [9], a genetic algorithm is used to extract both the intrinsic and extrinsic circuit elements of MEDFET and p-HEMT operating in 1–12 GHz and 1–50 GHz, respectively. In [10], artificial neural network has been used for temperature-dependent modeling of GaN HEMTs. In [11], a memetic differential evolution (DE) algorithm has been applied to obtain best fit optimized circuit elements for SSM of AlGaIn/GaN HEMT device. However, an appreciable amount of percentage error between the measured and calculated S-parameter values indicates some limitation of conventional algorithms that have the tendency to stick in local minima along with increased computation time resulting in suboptimal solution.

In this letter, a unique exploration strategy has been combined with evolutionary algorithms, such as PSO [12] and DE [13], to avoid the problem of local minima along with reduced computation time, resulting in robust optimization algorithms. In order to prove the effectiveness of the proposed methodology, optimization has been carried out using the above to obtain best fit intrinsic element values of a $2 \times 0.1 \times 50 \mu\text{m}^3$ GaN/Si HEMT and a $4 \times 0.1 \times 75 \mu\text{m}^3$ GaN/SiC HEMT, operating in a frequency range of 1–30 GHz at a particular bias point of interest.

II. SMALL-SIGNAL MODEL

The SSM considered for both the devices in the present investigation is shown in Fig. 1.

For both the devices, corresponding element values are extracted for initial data. These extracted data are then used

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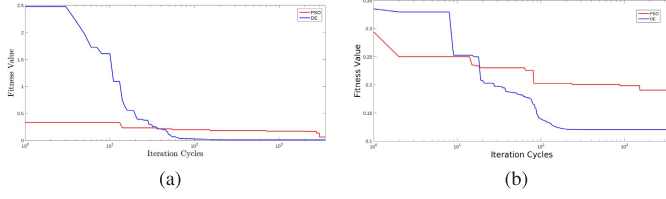


Fig. 2. Convergence profiles for optimizing intrinsic parameter values using PSO and DE for (a) GaN/Si HEMT and (b) GaN/SiC HEMT.

in conjunction with measured S-parameters for optimization using the above strategy. Accordingly, intrinsic S-parameter values are obtained by extracting the extrinsic parameters using a cold FET method and deembedding the same from measured S-parameters. The values obtained from experimental measurements for GaN/Si HEMT and GaN/SiC HEMT, respectively, have been used as a primary source of reference for optimization using PSO and DE. Intrinsic circuit elements that are being used to model the intrinsic SSM for both the devices are shown in Fig. 1.

III. PROPOSED METHOD

Every optimization algorithm depends upon a unique search space exploration strategy to obtain the optimum solution within the predefined search space. Optimization of intrinsic elements from SSM of different devices involves handling of very small nonnegative value. For example, the value of transit time τ is in fs ($\approx 10^{-12}$ s), resulting in very restricted modification in its value using random number-based conventional search space exploration strategy. Consequently, this may lead to local minima resulting in suboptimal convergence. In the proposed method of exploration, the intrinsic elements values are stripped off from their multipliers in the range of 10^{-12} or 10^{-15} while in the process of initialization and updating. While calculating S-parameter values in the optimization algorithm, the same intrinsic element values are multiplied with their respective multipliers. Every optimization algorithm works on a single-valued scalar function (called as fitness function) to achieve the optimization objective. In the present investigation, the fitness function F is as represented in

$$F = \left| \sum_{i=1}^N [(S_{11}(i)_m - S_{11}(i)_c) + 4 \times (S_{12}(i)_m - S_{12}(i)_c)] \right| + \left| \sum_{i=1}^N [(S_{21}(i)_m - S_{21}(i)_c) + (S_{22}(i)_m - S_{22}(i)_c)] \right| \quad (1)$$

where N represents the number of sample points considered in the operating frequency range, and $S_{11/12/21/22}(i)_m$ and $S_{11/12/21/22}(i)_c$ are the measured and calculated S-parameter values, respectively. Moreover, the values of S_{12} are much less than that of other S-parameter values. Consequently, in order to have a substantial influence of S_{12} on the error function, the second term in (1) is multiplied with a weight of 4. In order to validate the effectiveness of the proposed method, it is used with PSO [12] and DE [13] algorithms to obtain the optimized intrinsic element values for GaN/Si HEMT and GaN/SiC HEMT. Moreover, such an exploration strategy does

TABLE I
OPTIMIZED INTRINSIC ELEMENT VALUES WITH LB AND UB

Optimization method DE						
GaN/Si HEMT			GaN/SiC HEMT			
Element	LB	UB	Optimized Value	LB	UB	Optimized value
$R_{gs}(\Omega)$	1	100	1.219	1	100	4.62
$C_{gs}(fF)$	1	300	49.78	1	300	196.9
$C_{gd}(fF)$	1	100	10.68	1	100	15.14
$R_{gd}(\Omega)$	1	300	71.93	1	300	60.12
$g_m(mS)$	1	100	44.27	1	100	44.17
$\tau(ps)$	1	100	0.303	1	100	2.34
$R_{ds}(\Omega)$	1	2000	175.14	1	2000	897.92
$C_{ds}(fF)$	1	100	23.47	1	100	19.16
Optimization method PSO						
$R_{gs}(\Omega)$	1	100	2.62	1	100	8.22
$C_{gs}(fF)$	1	300	48.74	1	300	191.8
$C_{gd}(fF)$	1	100	10.79	1	100	14.9
$R_{gd}(\Omega)$	1	300	61.43	1	300	60.06
$g_m(mS)$	1	100	44.68	1	100	44.39
$\tau(ps)$	1	100	1.37	1	100	2.18
$R_{ds}(\Omega)$	1	2000	165.14	1	2000	954.94
$C_{ds}(fF)$	1	100	25.25	1	100	20.53

$[R_s = 1 \Omega, R_g = 1 \Omega, R_d = 0.5 \Omega, L_s = 0.1 pH, L_g = 55 pH$

$C_{pg} = 80 fF, C_{pd} = 52 fF, L_d = 41 pH$ for GaN/SiC]

$[R_s = 6.8 \Omega, R_g = 5.03 \Omega, R_d = 7.8 \Omega, L_s = 1.63 pH, L_g = 49 pH$

$C_{pg} = 128.7 fF, C_{pd} = 135.4 fF, L_d = 46.7 pH$ for GaN/Si]

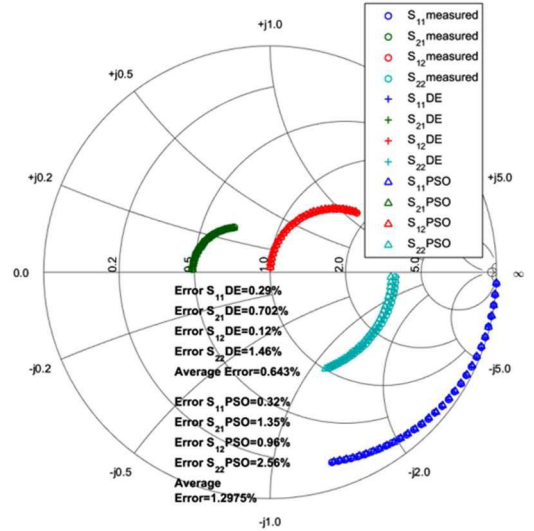


Fig. 3. Measured and optimized intrinsic S_{11} , S_{12} , S_{21} , and S_{22} of $2 \times 0.1 \times 50 \mu m^2$ GaN/Si HEMT biased at $V_{gs} = -2.25$ V and $V_{ds} = 5$ V from 1 to 30 GHz using PSO and DE.

not needs a prior knowledge of search source. The method can also be used for optimization of a temperature-dependent model by incorporating a thermal subnetwork in the equivalent circuit, which needs to be extracted separately.

IV. RESULTS

In the present investigation, optimization of intrinsic elements for SSM of GaN/Si HEMT and GaN/SiC HEMT operating in frequency range of 1–30 GHz using both DE and PSO

TABLE II
STATISTICAL PARAMETERS

Optimization Algorithm		DE	PSO
Device	Statistical Parameters	Value	Value
GaN/Si HEMT	Standard Deviation	0.0125	0.023
	Best Mean Cost	0.04251	0.1258
	Best fitness value	0.00948	0.06174
	Best Execution Time (sec.)	0.01595	628.23
GaN/SiC HEMT	Standard Deviation	0.00087	0.0671
	Best Mean Cost	0.1319	0.5425
	Best fitness value	0.1204	0.1907
	Best Execution Time (sec.)	2.917	647.74

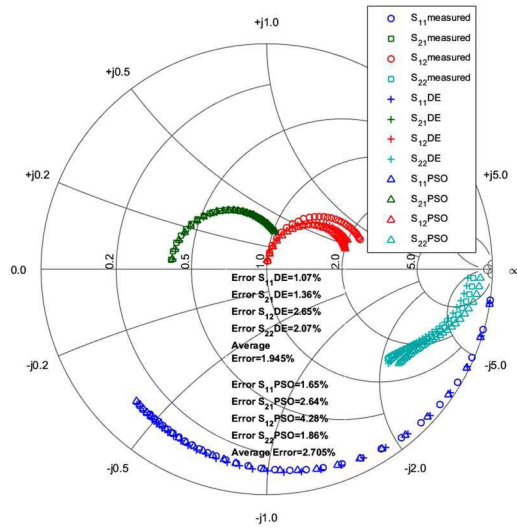


Fig. 4. Measured and optimized intrinsic S_{11} , S_{12} , S_{21} , and S_{22} of $4 \times 0.1 \times 75 \mu\text{m}^2$ GaN/SiC HEMT biased at $V_{gs} = -2.0$ V and $V_{ds} = 20$ V from 1 to 30 GHz using PSO and DE.

has been considered. The swarm size for both PSO and DE algorithms is kept at 35. GaN/Si HEMT is under bias condition of $V_{gs} = -2.25$ V and $V_{ds} = 5$ V, whereas for GaN/SiC $V_{gs} = -2$ V and $V_{ds} = 20$ V. Intrinsic small signal model for both devices are same resulting in the same set of optimization parameters. Consequently, C_{gs} , R_{gs} , R_{gd} , C_{gd} , g_m , C_{ds} , R_{ds} , and τ are considered as optimization parameters, whose upper bound (UB) and lower bound (LB) along with optimized values are summarized in Table I. The optimized values of intrinsic elements corresponds to best of the 25 independent trials for both PSO and DE algorithms with the above exploration strategy.

Fig. 2(a) and (b) shows the optimization curve for GaN/Si HEMT and GaN/SiC HEMT, respectively. Table II summarizes the corresponding statistical parameters for PSO and DE algorithms. Comparative S-parameter representation between the measured and modeled S-parameter values for GaN/Si HEMT and GaN/SiC HEMT are shown in Figs. 3 and 4, respectively. From the optimization curves and entries of Table II, it is observed that for both the devices DE outperforms PSO using the proposed search space exploration strategy in terms of faster and accurate convergence, which is more pronounced

in the case of the larger device operated at higher voltage. It is observed that, average error percentage for both devices are much less in the case of DE compared with PSO. Although DE in the present case is more accurate than PSO, but even the result obtained from PSO using the unique exploration strategy is improved from previous cases [6]. It is evident from the small standard deviation values of both PSO and DE that the algorithms used are stable.

V. CONCLUSION

Present investigation elaborates a general search space exploration strategy for evolutionary optimization algorithms used for compact SSM of microwave devices operating in the wide frequency range of 1 to 30 GHz under a particular bias condition. The effectiveness and generality of the proposed method is illustrated through optimization of intrinsic elements for SSM model of $2 \times 0.1 \times 50 \mu\text{m}^2$ GaN/Si HEMT and $4 \times 0.1 \times 75 \mu\text{m}^2$ GaN/SiC HEMT, using DE and PSO algorithms. Though DE outperforms PSO in all aspects, results obtained from PSO are also quite close to the measured S-parameters.

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